

Precision Agriculture & Targeted Intervention

PDDL / PDDL+

D5-V10
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The Challenge

A field does not pose one problem, rather it is many local decisions
It is required that the planner must avoid both missed treatment and unnecessary treatment.



Various local conditions

The model enforces the process ---> inspect → classify → targeted intervention

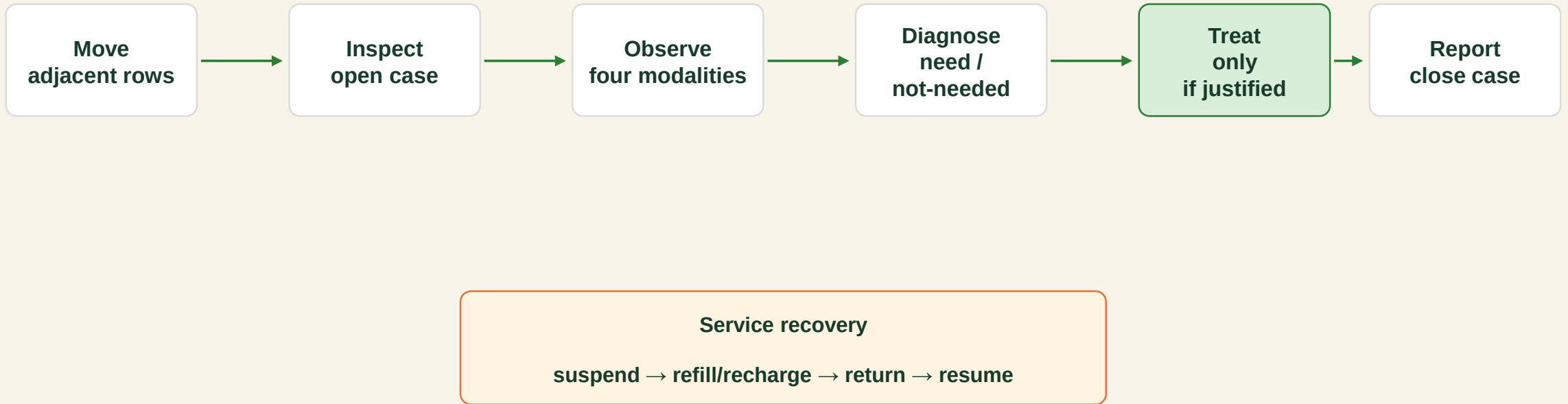
Modeling Requirements

	Inspect plots	→	travel / inspect / observe actions
	Classify conditions	→	diagnose need vs not-needed
	Target intervention	→	treatment gated by diagnosis
	Avoid harm	→	unsafe actions + damage events
	Resources matter	→	water, fertilizer, pesticide, energy

Treatments are not universally beneficial — evidence, resources, and safety need to be assessed.

Q1 · BASIC PDDL

Builds the symbolic decision
Classical PDDL captures the robot's deliberate task logic and resource/service recovery.



Assumption: crop conditions are symbolic and do not evolve while the robot moves.

Three Q1 problems demonstrate variable plans from one domain

The same domain handles single issue, multi-plot targeting, and service recovery.

P1 · localized issue

Non-depot scouting; healthy rows reported; one delayed-growth plot treated.



P2 · heterogeneous plots

Water only plot1; pesticide only plot2; fertilizer + support plot3; no treatment plot4.



P3 · service recovery

Diagnose need → insufficient resource → suspend → refill/recharge → return → treat.



Q2 · PDDL+

Hybrid behavior is built from actions, processes, and events
This pattern is used for travel, sensing, treatment, verification, and crop thresholds.

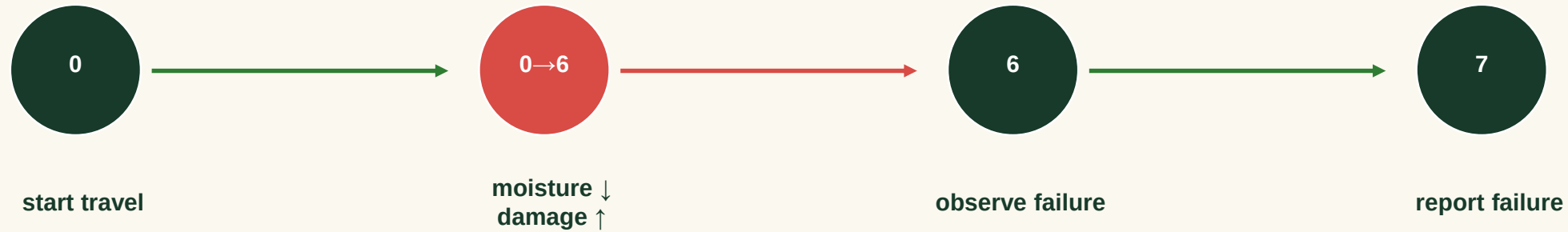


Example:

moisture decreases continuously →
threshold event creates moisture-deficient →
irrigation process restores moisture →
verification event enables safe report

Delayed intervention is not just late — it can become impossible

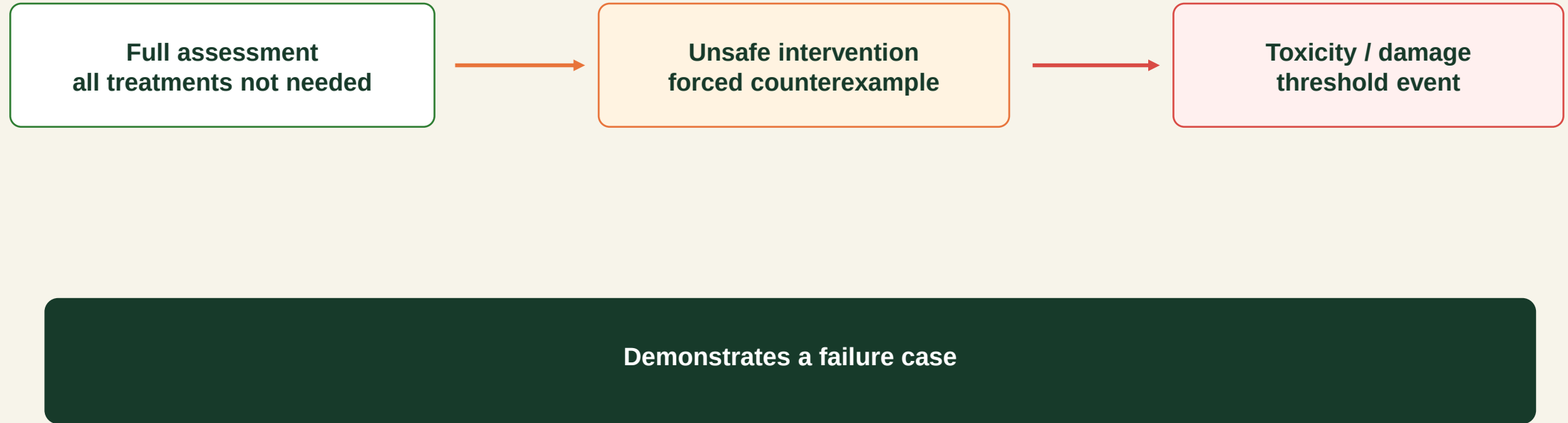
Problem 1 demonstrates failure emerging during travel.



Result: safe report becomes impossible if crop failure already occurred.

Wrong treatment is modeled as harmful, not just unnecessary

Problem 2 demonstrates treatment damage after an inappropriate/excessive intervention.



RECOVERY

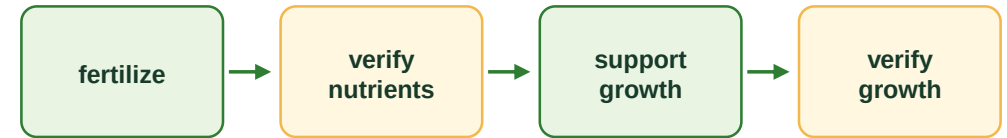
Problem 3 focuses on pest spread; Problem 4 focuses on staged nutrient/growth recovery.

P3 · pest spread



Dynamic spray runs from time 5→7, then pest verification enables safe report.

P4 · nutrient + growth



Staged ordering prevents growth support from being reported successful before nutrient recovery is verified.

Resource constraints

Q1 handles service recovery

Discrete symbolic behavior:
suspend case → refill/recharge → return → resume

Q2 handles hybrid crop dynamics

Continuous processes + threshold events:
condition evolution, dynamic treatment, verification

In Q2: if water is below the required dose, the robot reports resource limitation.

Assumptions

What is claimed by the model versus what is abstracted

- Closed world — facts not declared are false
- Deterministic sensing — no probability/noise
- Waypoint field — no low-level path dynamics
- Abstract resources — units, not litres/kg/joules

Q1
single issue + multi-plot + service recovery

Q2
processes, events, timing, wrong treatment

THANK YOU

